

Clinical Informatics Is Clinical Competence

A cross-specialty, competency-based approach to workplace assessment, progressive entrustment, and intentional sampling

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Disclosure

I am co-founder of Firefly Lab and equity holder and board member of the Firefly Foundation.

These conflicts of interest are unrelated to this presentation.



Informatics is no longer adjacent
to clinical care.
It is increasingly the medium
through which care is performed.

That makes informatics a cross-cutting clinical
competency—one that must be made visible, studied
systematically, and assessed with intention.



Eight Milestone Frameworks Already Train Informatics

In a direct review of eight representative ACGME Milestones 2.0 frameworks, the competency is explicit in three and embedded in five.

3

of 8 frameworks

EXPLICITLY NAME INFORMATICS / DIGITAL HEALTH

Pathology — SBP4: Informatics
Diagnostic Radiology — SBP8:
Informatics
Internal Medicine — PC6: Digital Health

5

of 8 frameworks

EMBED INFORMATICS WITHIN EXISTING MILESTONE LANGUAGE

General Surgery — SBP / PBLI	Family Medicine — SBP / PBLI
Emergency Medicine — PC / SBP	Obstetrics & Gynecology — SBP /
Anesthesiology — PC / SBP	PBLI
	Same clinical work; different label.

THE SHARED COMPETENCY LENS

Interpret data · Navigate workflow
Create information that supports care · Learn from longitudinal performance



Four Discrete Skills Make Informatics Assessable

Each is an enduring capability, expressed through specialty-specific clinical work—not a generic technology checklist.



Interpret clinical data

Extract a clinically meaningful, source-aware signal from heterogeneous data; recognize discordance and uncertainty.



Navigate digital workflow

Move safely across information systems and care transitions; identify information loss, workflow limits, and closed-loop needs.



Create information that supports care

Document and communicate a reliable clinical representation that enables safe downstream decisions and handoffs.



Learn from longitudinal performance

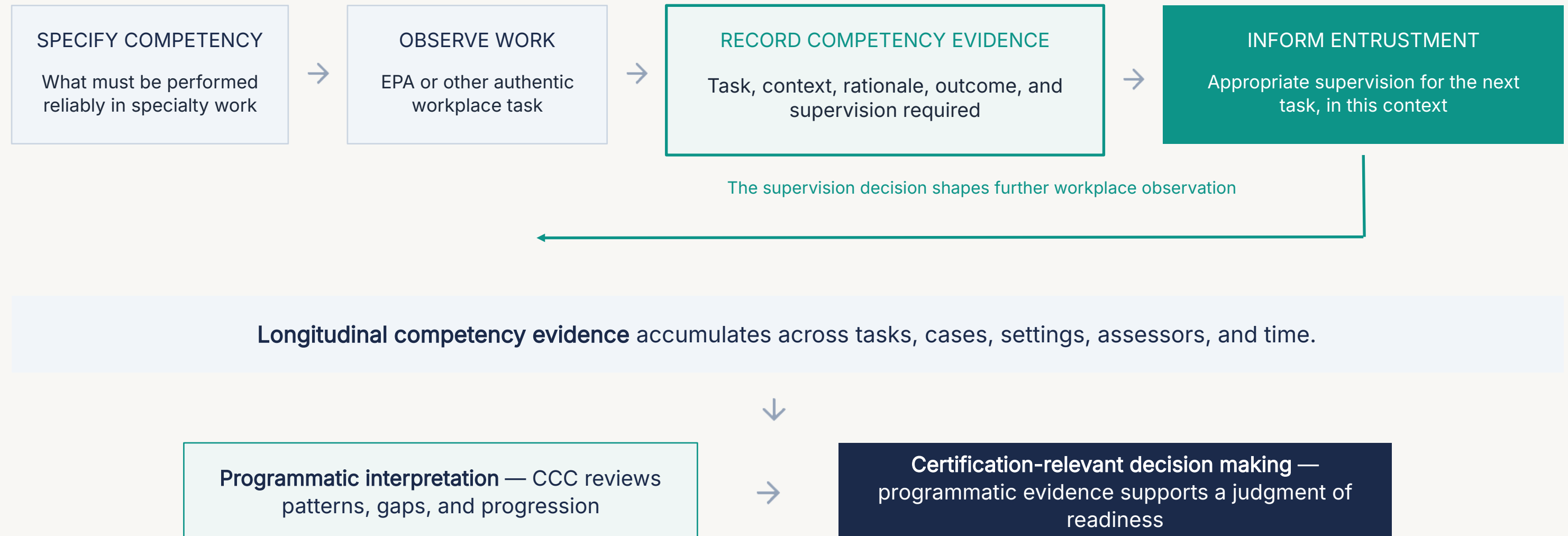
Use patient-, panel-, or system-level data to recognize variation, close care gaps, and improve clinical practice.

Competency-based design begins by naming the skill. Specialties then define the consequential clinical work in which that skill should be observed.



Workplace Evidence Informs Progressive Entrustment

Repeated clinical work makes competence visible; longitudinal evidence supports judgment—it is not a single-score proxy.



Competency is an enduring capability. Entrustment is a task- and context-specific supervision decision; both develop continuously through assessment and clinical work.



General Surgery: AI-Supported Reoperation Planning

A routine preoperative task becomes a high-value observation of embedded informatics competence.

ROUTINE RESIDENT WORK

Before a complex reoperation, the resident reconciles Epic, Care Everywhere, prior operative reports, and imaging. An illustrative AI-supported summary is reviewed—but verified against primary sources.

RETRIEVE

Cross-system record, imaging, and operative history

VERIFY

AI-supported synthesis against source records

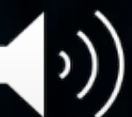
PLAN

Source-aware strategy, uncertainty, and handoff

WORKPLACE EVIDENCE

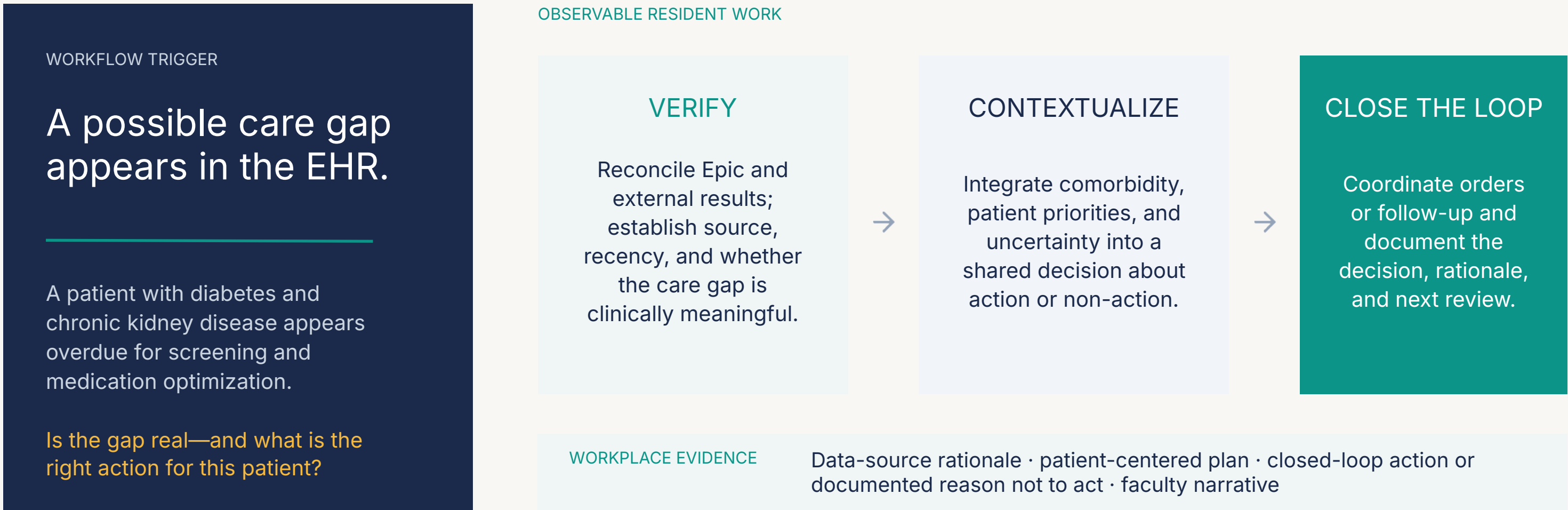
Source-aware problem representation · resolved or escalated discrepancy · verbal plan · faculty narrative on verification and supervision

Entrustment question: What supervision is appropriate for the next complex planning encounter with fragmented records and AI-supported input?



Internal Medicine: Closing a Care Gap With Longitudinal Data

A routine continuity-clinic task shows how EHR data becomes patient-centered clinical action.



Entrustment question: What supervision is appropriate for the resident’s next EHR-supported, longitudinal care-gap intervention?



One Assessment Grammar Can Serve Different Specialties

The shared structure is not the tool or EPA label; it is how a discrete clinical informatics skill becomes useful competency evidence.



THE SAME GRAMMAR, APPLIED TO SPECIALTY WORK

GENERAL SURGERY	Complex reoperation planning with fragmented records and AI-supported input	Verified problem representation + faculty narrative
INTERNAL MEDICINE	EHR-supported identification and closure of an actionable care gap	Source rationale + closed-loop plan + faculty narrative

An EPA is not the competency. It is a structured opportunity to observe a competency in consequential clinical work.



Intentional Sampling Determines Evidence Quality

Not every observation reduces uncertainty. The next assessment should be chosen for the information it is likely to add.

PASSIVE / CONVENIENCE SAMPLING

Observe what happens to be easy to assess.

Assessment concentrates in familiar tasks and among highly active faculty.
Complex, uncertain, or atypical work may remain unobserved.



Choose the observation that reduces uncertainty

UTILITY-AWARE SAMPLING

Observe where the evidence adds the most value.

Use workflow triggers to find authentic opportunities.
Use prior evidence to prioritize observations with the highest marginal information value.

UTILITY AS A DESIGN PRINCIPLE

$$\text{Utility} = \text{Reliability} \times \text{Validity} \times \text{Educational Impact} \times \text{Acceptability} \times \text{Cost}$$

Intentional assessment is not more assessment. It is a program of observation designed to produce credible evidence for progressive entrustment.



The Surgical Literature Shows That the Signal Is Designed

Evidence used for competence and entrustment reflects the assessment system—not only the learner's performance.

60.8%

FACULTY PARTICIPATION CONCENTRATION

of EPA submissions came from the most active 10% of faculty across 46 institutions.

52.8%

EPA-OPPORTUNITY CONCENTRATION

of 17,245 assessments involved only 4 of 18 EPA types across 37 institutions.

2.72 → 2.65

SAMPLING-SOURCE DIFFERENCE

Mean research rating, manual versus EHR-triggered assessments; **p = 0.005.**

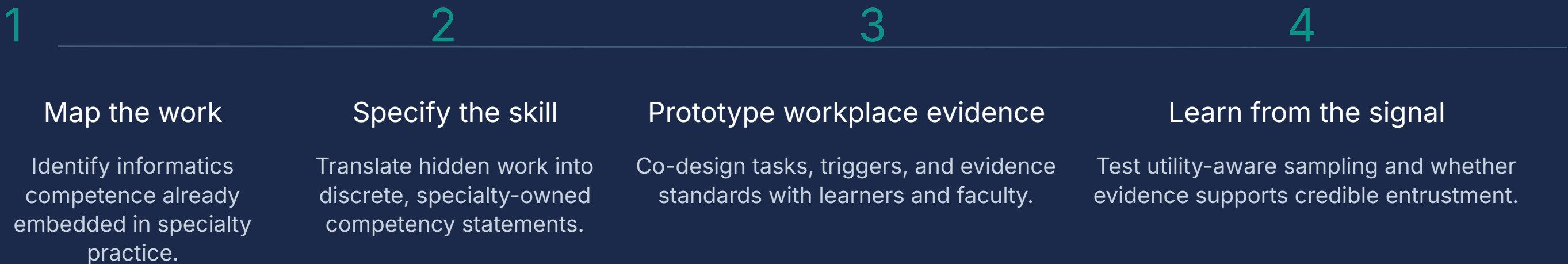
Workplace assessment does not fail because it is workplace assessment.
It fails when taxonomy, sampling, and implementation are left to chance.

For clinical informatics, build the signal intentionally—from specialty work, workflow triggers, and information value.



A Cross-Specialty Collaboration Can Build This Deliberately

The opportunity is to connect competency already embedded in clinical work to credible evidence, progressive entrustment, and certification.



Informatics is everywhere in clinical work.
Its assessment should be intentional.

The next step is a cross-specialty collaboration to build, test, and refine this evidence system together.



Selected References

Cross-specialty milestone mapping, competency-based assessment foundations, and the speaker’s surgical assessment research

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